Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/29/2021
PI Name:	Peters, Brian Ph.D.		
Project Title:	Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	07/27/2020	**End Date:**	09/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reschke, Millard Ph.D. (NASA Johnson Space Center) Wood, Scott Ph.D. (NASA Johnson Space Center) Rosenberg, Marissa Ph.D. (KBR/NASA Johnson Space Center) Caldwell, Erin M.S. (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation and augmentation of "Orion Crew Performance and Safety Immediately Following Postflight Water Landings: Emergency Egress (PI Reschke)" with new Principal Investigator Dr. Brian Peters. The decision to return to water landings in the Orion capsule presents significant safety issues that must be addressed. First, based on current models, it is anticipated that stable 2 capsule attitude upon splash down will occur more that 51% of the time and remain in that position for up to 20 min. Second, currently 100% of returning International Space Station (ISS) crewmembers (both Russian and US Orbital Segment (USOS)) have landing sickness, the severity of which is expected to increase due to wave motion coupled with exposure to a closed visual environment (cannot see a horizon). Third, strength, while compromised, is currently not believed to be an issue for emergency egress in long duration crewmembers (20+ days). However, crewmembers are vestibular and sensorimotor challenged to the point where coordinated walking, rising from a seat, or standing is difficult. Fourth, 100% of returning Soyuz crews when questioned about an unaided emergency egress from the Orion in the water have said that it will be catastrophic. The goal of the augmentation effort is to identify the optimal features of a device that visually presents gravitational reference cues to aid stability and re-adaptation that can be deployed in a capsule or aboard the recovery ship. The first specific aim is to enhance the existing capsule sensory-condition simulator (used in the existing Omnibus project) to include instrumented hand-holds, a compliant support surface and galvanic vestibular stimulation (GVS). Subjects in the previous investigation were destabilized by the wave motions but easily able to remain standing without support. They effectively used vestibular and proprioceptive cues to make the necessary adjustments to stay upright. The addition of the vestibular and proprioceptive challenges could easily increase the risk of falling. To mitigate this risk, hand holds will be added inside the simulator. The handles will also be instrumented with load cells to measure the amount of force imparted to them. Simple subject instructions, to minimize this force throughout each trial, will allow measures of the magnitude of the force to serve as the primary dependent variable when comparing test conditions. The second specific aim is to develop and test multiple presentation methods of an Earth-fixed visual reference. During this aim we will explore different visual aid options, both electrical and physical, to determine which can provide the best stabilizing effect. Special consideration is given to options that would be viable within the capsule, but it is important to remember that the stability challenges will also be present in an internal room on the recovery ship. Vertical and horizontal line projections will be compared as will the use of points of light instead of lines. Projected references could be problematic in the capsule so mechanical and custom-controlled light displays will also be explored. These conditions will be compared using a repeated measures design. The learning effects that complicated the interpretation of the previously collected data will be controlled by both a balanced test order and the inclusion of a control condition performed at the beginning and end of each test session. Perfectly balanced condition orders can be obtained for the comparison of either 3 or 4 conditions, using 24 subjects. Two test sessions will be conducted during which two sequential 60 second trials will be collected for each condition. The first of the trials will be considered to be a training period for each condition and not analyzed. The results of the comparisons and recommendations for best characteristics to be used in the implementation of the countermeasure will be provided in a final report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal effort sets out to develop a test that could be performed on returning crewmembers to better quantify astronauts' ability to egress the capsule after a water landing. A secondary goal is to compare the performance between a condition during which an Earth-fixed visual reference was projected onto the walls of the enclosure and another without the visual cue. An Earth-fixed visual reference has applications for other motion platforms, including those on air, land, or sea.
Task Progress:	Progress on this project has been significantly impacted by restrictions imposed as a result of COVD-19. Funding was initiated in October of 2020 and to date, work on this project has not been approved for onsite activity. Progress on some software development aspects of the project was also impacted when a software development computer became nonfunctional. Replacement of the computer was slowed by COVID-related delays. Continuation of work on a computer-controlled visual display that will serve as one of the Earth-fixed visual reference options to be tested has recently begun. Progress in the design of modifications to the motion platform-mounted test enclosure was made possible by equipping one of the lab engineers with SolidWorks, a computer-aided design (CAD) software package. Computer models of the test hardware components were developed/modified and options for the implementation of changes to the enclosure were developed. These parts have been ordered and assembly can begin when approval for onsite activity for this project is granted.
Bibliography Type:	Description: (Last Updated: 07/29/2021)
Abstracts for Journals and Proceedings	Peters BT, Caldwell EE, Wood SJ, Reschke MF. "Developing an Earth-fixed visual reference to aid stability, readaptation and egress after a water landing." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021